

PROBAL SEN

Kolkata, West Bengal

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Probal Sen

Probal-Sen

Education

University Institute of Technology, Burdwan University

2023 – Present

B.E. in Information Technology — CGPA: 9.23

Burdwan, WB

Ashoknagar Boys' Secondary School (H.S.), WBCHSE

2020 – 2022

Class XII – PCM — 90%

Ashoknagar, WB

Skills

Core: Data Structures & Algorithms, Operating Systems, OOP, DBMS

Languages: C++, C, Java, Python

Web: HTML5, CSS3, JavaScript, MongoDB, Node.js

Data Science & ML: Pandas, NumPy, Matplotlib, Scikit-learn

Tools: VS Code, GitHub, Power BI

Soft Skills: Teamwork, Problem-Solving, Adaptability, Communication

Experience

AI Intern – Habitat Lens Private Limited

Oct 2025 – Dec 2025

Image Annotation – Computer Vision (84 Hours)

Contributed to the Handwritten Notes Understanding Challenge by labeling handwritten text images using Python-based **LabelImg**. Created bounding box annotations for object detection, ensured dataset accuracy, and supported preprocessing for AI/ML model training.

AICTE Internship – Edunet Foundation

Sept 2025 – Oct 2025

Intern – Artificial Intelligence & Cloud Technology

Completed a 4-week internship under the AICTE SkillsBuild Program. Gained hands-on experience in AI and Cloud through project-based learning, collaborated with mentors on real-world solutions, and attended expert masterclasses.

Projects

UIT Burdwan University Inter-Department Football Tournament Web App

Dec 2025

Next.js (TypeScript), Node.js, Express.js, MongoDB, Socket.io, JWT

Developed a full-stack tournament management system with admin-only JWT authentication and public dashboards for fixtures, live matches, and standings. Implemented real-time score updates using Socket.io, automated points table calculation, and RESTful APIs for managing teams, players, and matches. Designed a scalable architecture with MongoDB and ensured seamless frontend-backend communication.

Solar Energy Prediction using Machine Learning

Nov 2025

Python, Pandas, NumPy, Matplotlib, Scikit-learn

Built a machine learning model to predict solar energy output (kWh) using environmental and weather features such as irradiance, temperature, and cloud cover. Applied data preprocessing, feature engineering, and trained a Random Forest Regressor achieving high accuracy ($R^2 = 0.937$, RMSE = 1.23 kWh). Visualized model performance using prediction and residual analysis plots.

SoulSupport – AI Mental Wellness Companion

Oct 2025

Python, Flask, HTML, CSS, JavaScript, Google Gemini AI

Developed an AI-powered web application with a sentiment-aware chatbot for personalized emotional support. Integrated mood tracking features and designed a responsive user interface to enhance user engagement and accessibility.

Certifications

JavaScript, Python – HackerRank

Git Training – Simplilearn

Postman API Fundamentals – Postman